Vamsi Krishna

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PROFESSIONAL SUMMARY

Data Engineer with 4+ years building cloud-native data platforms across healthcare, finance, banking, insurance, and e-commerce, with strong delivery experience in the Microsoft data ecosystem.
Built batch and streaming pipelines using **Azure Data Factory**, **Azure Databricks**, **AWS Glue**, **Spark**, **Kafka**, and **GCP Dataflow**, supporting curated analytics, telemetry, and regulatory workloads.
Designed lakehouse and warehouse patterns on **ADLS Gen2**, **Amazon S3**, **Delta Lake**, **Azure Synapse Analytics**, **BigQuery**, and **Amazon Redshift**, including Medallion layers and star schema reporting models.
Implemented governance and security with **Azure Purview**, **AWS Glue Data Catalog**, **IAM**, **RBAC**, **KMS**, and **Azure Key Vault**, aligned to HIPAA, SOC2, and GDPR-driven controls.
Supported production operations with **Airflow**, **Azure DevOps**, **Azure Monitor**, **Log Analytics**, **CloudWatch**, and **Step Functions**, including 50+ workflows and environments that reached 98% availability in production.

PROFESSIONAL SKILLS

• **Programming Languages & Scripting:** Python, PySpark, Scala, Java, SQL, T-SQL, PL/pgSQL, PL/SQL, Bash
• **Data Processing :** Apache Spark, Spark SQL, Databricks, AWS EMR, Azure HDInsight, GCP Dataflow, Apache Beam, Delta Lake, Medallion architecture, dimensional modeling, star schemas
• **Data Ingestion & Integration:** Azure Data Factory, AWS Glue, Kafka, Kinesis, Azure Event Hubs, Pub/Sub, SSIS, Talend, Informatica, Matillion
• **Warehouses & Analytics Engines:** Azure Synapse Analytics, Amazon Redshift, Snowflake, BigQuery, Spark SQL
• **Data Lake & Governance:** ADLS Gen2, Amazon S3, Delta Lake lakehouse, Azure Purview, AWS Glue Data Catalog, dbt, lineage, sensitivity labeling, PII classification
• **Security & Compliance:** IAM, KMS, RBAC, PrivateLink, VPC Endpoints, Azure Key Vault, encryption at rest and in transit, HIPAA, SOC2, GDPR
• **Observability & Reliability:** Azure Monitor, Log Analytics, Amazon CloudWatch, alerting, runbook-driven triage, Airflow monitoring, schema drift handling, validation checkpoints
• **CI/CD & Release Engineering:** Azure DevOps Pipelines, Jenkins, GitHub, GitLab, AWS CodeBuild, CI/CD, code reviews, release promotion
• **Infrastructure as Code (IaC):** Terraform, CloudFormation, ARM Templates
• **Cost & Performance:** partitioning, clustering, indexing, Synapse Dedicated SQL Pool tuning, storage lifecycle policies, blob tiering, query optimization
• **SDLC & Collaboration:** Agile delivery, Jira, Confluence, stakeholder requirements, documentation, mentorship, code reviews, QA and release coordination

PROFESSIONAL EXPERIENCE

Ryan Specialty — Azure Data Engineer | Chicago, IL | Aug 2024 – Present
• Built a **Delta Lake** lakehouse on **ADLS Gen2** with Bronze, Silver, Gold layers, standardized raw landing and curated consumer datasets for claims, clinical, and billing analytics.
• Created parameterized **Azure Data Factory** pipelines with dynamic datasets, triggers, and metadata-driven routing, onboarded repeatable healthcare feeds with less manual rework across environments.
• Engineered **PySpark** transformations in **Azure Databricks** to clean, enrich, and unify semi-structured healthcare data used for actuarial and medical cost analysis.
• Integrated **Azure Event Hubs** and **Stream Analytics** for EHR, telehealth, and sensor streams, kept low-latency routing stable during bursty arrivals.
• Added schema drift detection, validation checkpoints, and notebook-level quality rules, reduced downstream breakage when upstream providers changed payloads and file layouts.
• Modeled reporting datasets in **Azure Synapse Analytics** using star schemas, tuned indexing and distributed query patterns to support BI workloads.
• Tuned **Synapse Dedicated SQL Pools** for partitioned joins and predictable runtimes, supported actuarial reporting peaks without destabilizing scheduled loads.
• Secured regulated data paths with **Azure Key Vault**, **RBAC**, and private endpoints, enforced least-privilege access aligned to HIPAA and SOC2 controls.
• Registered assets in **Azure Purview** with lineage, sensitivity labels, and PII classification, improved audit readiness across **ADF**, **Databricks**, and **Synapse** flows.
• Automated environment provisioning with **Terraform** and **ARM Templates**, kept dev, test, and prod builds consistent across storage, networking, and data services.
• Deployed **Azure DevOps** YAML pipelines for **ADF**, **Databricks**, and **Synapse** artifacts, tightened release promotion and rollback discipline across environments.
• Implemented blob tiering and lifecycle policies for archival datasets, reduced long-term storage spend while preserving retention requirements.

HCL Technologies — AWS Data Engineer | USA | Dec 2023 – Jul 2024
• Delivered enterprise ETL workflows in **Azure Data Factory** for structured and semi-structured claims, policy, and financial datasets across healthcare and insurance domains.
• Built **PySpark** transformations in **Azure Databricks** for cleansing and enrichment, handled distributed processing for large-scale batch workloads.
• Implemented **Delta Lake** Medallion Bronze, Silver, Gold layers on **ADLS Gen2**, supported schema evolution, lifecycle tracking,

and reusable curated datasets.
• Engineered streaming ingestion with **Azure Event Hubs**, **Stream Analytics**, and **Spark Structured Streaming**, supported low-latency telemetry and transaction routing.
• Integrated **Azure Purview** with **ADF**, **Databricks**, and **Synapse**, strengthened lineage and governance for audit trails and compliance reporting.
• Monitored pipeline health with **Azure Monitor** and **Log Analytics**, maintained 98% production availability with actionable alerting and triage dashboards.
• Deployed **Azure DevOps** YAML pipelines and **Terraform** for repeatable promotion of **ADF**, **Databricks**, and **Synapse** artifacts across environments.
• Secured data access with **Azure Key Vault**, **RBAC**, private endpoints, and VNET integration, met HIPAA and SOC2 security expectations for regulated workloads.
• Built notebook frameworks for schema drift handling, late-arriving updates, and data quality checks, reduced rerun effort during production backfills.
• Mentored junior engineers on governance and CI/CD standards, improved sprint delivery consistency and code review quality.

Amazon — GCP Data Engineer | Chennai, India | May 2022 – Jul 2023
• Built **Kafka** and **Spark Streaming** pipelines for high-volume event processing, enabled real-time data movement across internal systems.
• Designed **GCP Dataflow** pipelines with **Apache Beam** for structured and semi-structured datasets, standardized transforms across multiple producer feeds.
• Orchestrated 50+ **Airflow** and **Databricks** workflows, implemented reusable DAG templates, dependency controls, and rerun paths for operational recovery.
• Developed **BigQuery** ELT frameworks with SQL, partitioning, clustering, and stored procedures, improved performance for analytics consumers.
• Integrated **Pub/Sub**, **GCS**, **Dataflow**, and **BigQuery** for event-driven ingestion, improved decoupling between ingestion and serving layers.
• Implemented schema evolution handling across batch and streaming jobs, reduced failures caused by frequent upstream payload changes.
• Authored **Terraform** modules for **BigQuery**, **Pub/Sub**, and service accounts, improved repeatability of environment provisioning.
• Added retry logic, checkpointing, dead letter queues, and fallback paths, reduced disruption during transient broker and consumer failures.

Bank of America — Data Engineer | Chennai, India | Mar 2021 – Apr 2022
• Built **Azure Databricks** pipelines with **PySpark** and **Spark SQL** to process terabyte-scale financial and regulatory datasets for risk and compliance reporting.
• Delivered end-to-end ETL across **Azure Data Lake**, **Databricks**, and SQL systems, supported reporting schedules tied to month-end cycles.
• Leveraged **Hive** and **HBase** for aggregation of regulatory metrics and time-series datasets used by finance and compliance teams.
• Implemented governance and lineage tracking in **Azure Data Lake**, improved audit-ready transparency for regulated datasets.
• Secured ingestion with **RBAC**, service principals, and **Azure Key Vault**, protected sensitive financial data access paths.
• Built validation layers for schema checks, data quality rules, and exception reporting, reduced downstream reprocessing caused by malformed inputs.
• Created **Delta Lake** tables for incremental loads, ACID behavior, and time travel analysis supporting reconciliation and investigation work.
• Deployed containerized transformation services with **Docker** and **Kubernetes**, improved portability and scale across environments.
• Built observability dashboards with **Azure Monitor** and **Log Analytics**, improved failure visibility and faster routing of fixes during incidents.

PROJECTS

• Healthcare Lakehouse on Azure — Data Engineer, **ADLS Gen2**, **Delta Lake**, **Azure Data Factory**, **Azure Databricks**, **PySpark**, **Azure Synapse**, **Azure Purview**, **Azure Key Vault**, **Azure DevOps**, **Terraform**, built curated analytics layers for claims and clinical domains, enforced schema drift detection and validation checkpoints, implemented lineage and PII classification for audit readiness, supported reliable delivery with 98% production availability.
• Real-Time Event Processing on GCP — Data Engineer, **Kafka**, **Spark Streaming**, **Pub/Sub**, **GCP Dataflow**, **Apache Beam**, **BigQuery**, **Airflow**, **Terraform**, designed event-driven ingestion with schema evolution handling, checkpointing, and dead letter queues, added end-to-end observability for low-latency datasets used by analytics and ML feature consumers.

EDUCATION

• Master of Science, Data Science, University of New Haven — 2025

CERTIFICATIONS

• Microsoft Certified, Azure Data Engineer Associate
• Databricks Certified Data Engineer Associate